

Phase 5 — Cart & Checkout Implementation Completed

All features, stores, services, components, and verification gates for **Phase 5 (Cart & Checkout)** are implemented and verified. The full completion report is available in `phase_5_completion_report.md`.

Verification Summary

Gate	Command	Result
TypeScript Compilation	<code>npx tsc --noEmit</code>	PASS (0 errors)
ESLint Quality	<code>npm run lint</code>	PASS (0 errors)
SCSS Stylelint	<code>npm run lint:style</code>	PASS (0 errors)
Unit Test Suite	<code>npm test -- --watch=false --no-progress</code>	PASS (250/250 SUCCESS)
Production Build	<code>npm run build</code>	PASS (0 errors)
Backend Purity	<code>git status --porcelain Back-end/</code>	PASS (Completely untouched)

Key Deliverables & Architecture

1. Guest Session & HTTP Interception:
- `guest-session.service.ts`: Client-side UUID session generator with secure local persistence and lifecycle cleanup on user authentication.
 - `guest-session.interceptor.ts`: Injects `x-guest-token` header onto all unauthenticated cart and checkout HTTP requests.
2. State Management:
- `cart.store.ts`: Signal-driven cart store featuring optimistic mutations, concurrency version tracking (`expectedVersion`), 409 conflict detection, and cart merge conflict resolution.
 - `checkout.store.ts`: Multi-step checkout wizard (`information` -> `payment` -> `review` -> `confirmation`), dynamic shipping estimation by governorate, free pickup, coupon validation (`AZHARI10`), manual payment methods (Wallets, InstaPay, Bank Transfer, COD), Cloudinary direct upload configs, and idempotency key safety.
3. UI Components & Pages:
- `cart.component.ts`: Dedicated `/cart` page with quantity steppers, item removal, clear cart confirmation, and stale cart alerts.
 - `checkout.component.ts`: Multi-step checkout flow with order summary breakdown, order placement with `Idempotency-Key` , reference number copy tool, and screenshot upload dropzone for manual payment proof.
 - `storefront-shell.component.ts`: Header cart badge counter and interactive mini-cart dropdown.
 - `product-detail.component.ts`: Direct integration to add items to cart with live quantity controls.

PHASE 5 COMPLETE.
Execution stopped intentionally at Phase 5.
No Phase 6+ implementation was performed.